

```
--Week 3 Assignment - Sales Analysis

--Sample Superstore Dataset consist of :
--Rows: 9994
--Columns: 21
```

```
CREATE TABLE superstore (
    row_id INTEGER,
    order_id VARCHAR(20),
    order_date TEXT,
    ship_date TEXT,
    ship_mode VARCHAR(50),
    customer_id VARCHAR(20),
    customer_name VARCHAR(150),
    segment VARCHAR(50),
    country VARCHAR(100),
    city VARCHAR(100),
    state VARCHAR(100),
    postal_code VARCHAR(20),
    region VARCHAR(50),
    product_id VARCHAR(50),
    category VARCHAR(50),
    sub_category VARCHAR(50),
    product_name TEXT,
    sales NUMERIC,
    quantity INTEGER,
    discount NUMERIC,
    profit NUMERIC
);
```

```
--Customers table
CREATE TABLE customers AS
SELECT DISTINCT
    customer_id,
    customer_name,
    segment
FROM superstore;
```

```
--Product Table
CREATE TABLE products AS
SELECT DISTINCT
    product_id,
    product_name,
    category,
    sub_category
FROM superstore;
```

```
--order table
CREATE TABLE orders AS
SELECT DISTINCT
    order_id,
    order_date,
    ship_date,
    ship_mode,
    customer_id,
    product_id,
```

```

    sales,
    quantity,
    discount,
    profit
FROM superstore;

```

--Queries

--1. Find all orders where sales are greater than the average sales.

(Subquery)

```

SELECT *
FROM superstore
WHERE sales >
(
    SELECT AVG(sales)
    FROM superstore
);

```

Output:

Open File Put Messages Notifications Ctrl O SQL Showing rows: 1 to 1000 Page No: 1 of 3									
	row_id integer	order_id character varying (20)	order_date text	ship_date text	ship_mode character varying (50)	customer_id character varying (20)	customer_name character varying (150)	segment character varying (20)	
1	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	
4	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
5	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
6	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	
7	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	
8	17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Kriz	Consumer	
9	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	
10	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	
11	36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	
12	39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	
13	41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	
14	55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	
Total rows: 2360 Query complete 00:00:01.667 CRLF Ln 74, Col 3									

Insight: Only a subset of orders generated sales above the overall average sales value.

These orders represent high-value transactions and contribute significantly to total revenue.

--2. Find the highest sales order for each customer. (Subquery)

```

SELECT *
FROM superstore s
WHERE sales =
(
    SELECT MAX(sales)
    FROM superstore
    WHERE customer_id = s.customer_id
);

```

Output:

Data Output Messages Notifications								
Showing rows: 1 to 795 Page No: 1 of 1								
	row_id integer	order_id character varying (20)	order_date text	ship_date text	ship_mode character varying (50)	customer_id character varying (20)	customer_name character varying (150)	segment character varyi
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
2	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
3	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
4	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer
5	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer
6	36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate
7	39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office
8	55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate
9	58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer
10	68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate
11	110	CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	Pete Armstrong	Home Office
12	146	CA-2015-110744	9/7/2015	9/12/2015	Standard Class	HA-14920	Helen Andreada	Consumer
13	150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate
Total rows: 795 Query complete 00:00:31.875 CRLF Ln 84, Col 3								

Insight: Each customer has at least one order that contributes the most to their spending.

Identifying these orders helps understand customer purchasing behavior and peak transaction values.

--3. Calculate total sales for each customer. (CTE)

WITH customer_sales AS

```
(
    SELECT
        customer_id,
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_id, customer_name
)
```

```
SELECT *
FROM customer_sales;
```

Output:

Data Output Messages Notifications			
Showing rows: 1 to 793			
	customer_id character varying (20)	customer_name character varying (150)	total_sales numeric
1	SC-20440	Shaun Chance	1874.998
2	MH-17785	Maya Herman	2203.778
3	CK-12595	Clytie Kelty	2372.750
4	DW-13195	David Wiener	1080.751
5	CL-11890	Carl Ludwig	1262.014
6	VT-21700	Valerie Takahito	1736.596
7	MH-18115	Mick Hernandez	5503.093
8	HM-14860	Harry Marie	8236.7648
9	ME-18010	Michelle Ellison	1256.944
10	KD-16345	Katherine Ducich	1888.964
11	BF-11005	Barry Franz	1333.880
12	EM-13960	Eric Murdock	933.704
13	TS-21085	Thais Sissman	4.833
14	EH-14125	Eugene Hildebrand	3082.653
Total rows: 793		Query complete 00:00:00.432	

Insight: Total sales vary considerably among customers.
This aggregation helps identify high-value customers and measure customer contribution to overall revenue.

```
--4. Find customers whose total sales are above average. (CTE + Subquery)
WITH customer_sales AS
(
    SELECT
        customer_id,
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_id, customer_name
)

SELECT *
FROM customer_sales
WHERE total_sales >
(
    SELECT AVG(total_sales)
    FROM customer_sales
);
```

Output:

Data Output Messages Notifications			
Showing rows: 1 to 294			
	customer_id character varying (20)	customer_name character varying (150)	total_sales numeric
1	MH-18115	Mick Hernandez	5503.093
2	HM-14860	Harry Marie	8236.7648
3	EH-14125	Eugene Hildebrand	3082.653
4	SF-20200	Sarah Foster	3422.786
5	MG-18145	Mike Gockenbach	3061.543
6	BM-11650	Brian Moss	7294.185
7	DB-13555	Dorothy Badders	3908.8028
8	JW-15220	Jane Waco	7721.714
9	DG-13300	Deirdre Greer	3195.818
10	TB-21400	Tom Boeckenhauer	9133.990
11	SF-20965	Sylvia Foulston	3336.535
12	IL-15100	Ivan Liston	5040.736
13	DR-12880	Dan Reichenbach	6528.034
14	GB-14575	Giulietta Banti	4716.288
Total rows: 294		Query complete 00:00:00.553	

Insight: Only a limited number of customers generate sales above the average customer spending level. These customers can be considered key revenue contributors and may be targeted for retention strategies.

--5. Rank all customers based on total sales. (Window Function)

WITH customer_sales AS

```
(
    SELECT
        customer_id,
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_id, customer_name
)
```

```
SELECT
    customer_id,
    customer_name,
    total_sales,
    RANK() OVER(ORDER BY total_sales DESC) AS sales_rank
FROM customer_sales;
```

Output:

Data Output Messages Notifications				
Showing rows: 1 to 793				
	customer_id character varying (20)	customer_name character varying (150)	total_sales numeric	sales_rank bigint
1	SM-20320	Sean Miller	25043.050	1
2	TC-20980	Tamara Chand	19052.218	2
3	RB-19360	Raymond Buch	15117.339	3
4	TA-21385	Tom Ashbrook	14595.620	4
5	AB-10105	Adrian Barton	14473.571	5
6	KL-16645	Ken Lonsdale	14175.229	6
7	SC-20095	Sanjit Chand	14142.334	7
8	HL-15040	Hunter Lopez	12873.298	8
9	SE-20110	Sanjit Engle	12209.438	9
10	CC-12370	Christopher Conant	12129.072	10
11	TS-21370	Todd Sumrall	11891.751	11
12	GT-14710	Greg Tran	11820.120	12
13	BM-11140	Becky Martin	11789.630	13
14	SV-20365	Seth Vernon	11470.950	14
15	CJ-12010	Caroline Jumper	11164.974	15
Total rows: 793		Query complete 00:00:00.564		

Insight: Customer ranking highlights the highest revenue-generating customers.

The ranking can be used for loyalty programs, customer segmentation, and business decision-making.

--6. Assign row numbers to each order within a customer. (Window Function + PARTITION BY)

```
SELECT
    customer_id,
    customer_name,
    order_id,
    sales,
    ROW_NUMBER()
    OVER(
        PARTITION BY customer_id
        ORDER BY sales DESC
    ) AS row_num
FROM superstore;
```

Output:

Data Output Messages Notifications					
Showing rows: 1 to 1000					
	customer_id character varying (20)	customer_name character varying (150)	order_id character varying (20)	sales numeric	row_num bigint
1	AA-10315	Alex Avila	CA-2016-103982	3930.072	1
2	AA-10315	Alex Avila	CA-2014-128055	673.568	2
3	AA-10315	Alex Avila	CA-2016-103982	431.976	3
4	AA-10315	Alex Avila	CA-2017-147039	362.94	4
5	AA-10315	Alex Avila	CA-2014-128055	52.98	5
6	AA-10315	Alex Avila	CA-2016-103982	41.72	6
7	AA-10315	Alex Avila	CA-2015-121391	26.96	7
8	AA-10315	Alex Avila	CA-2014-138100	14.94	8
9	AA-10315	Alex Avila	CA-2014-138100	14.56	9
10	AA-10315	Alex Avila	CA-2017-147039	11.54	10
11	AA-10315	Alex Avila	CA-2016-103982	2.304	11
12	AA-10375	Allen Arnold	CA-2016-131065	499.98	1
13	AA-10375	Allen Arnold	CA-2015-140921	149.97	2
14	AA-10375	Allen Arnold	CA-2017-100230	116.98	3
15	AA-10375	Allen Arnold	CA-2015-114503	84.96	4
Total rows: 9994		Query complete 00:00:00.317			

Insight: Orders are sequentially numbered for each customer based on sales value.
This makes it easier to identify a customer's highest, second-highest, and subsequent orders.

--7. Display top 3 customers based on total sales. (Window Function)

WITH customer_sales AS

```
(
    SELECT
        customer_id,
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_id, customer_name
)
```

```
SELECT *
FROM
(
    SELECT
        customer_name,
        total_sales,
        RANK() OVER(ORDER BY total_sales DESC) AS rank_no
    FROM customer_sales
)
```

```
) x
WHERE rank_no <= 3;
```

Output:

	customer_name character varying (150) 🔒	total_sales numeric 🔒	rank_no bigint 🔒
1	Sean Miller	25043.050	1
2	Tamara Chand	19052.218	2
3	Raymond Buch	15117.339	3

Insight: The top three customers contribute a substantial portion of total sales.

These customers represent the most valuable segment of the customer base.

```
--Final Combined Query(Use JOIN + CTE + Window Function together)
WITH customer_sales AS
(
    SELECT
        c.customer_id,
        c.customer_name,
        SUM(o.sales) AS total_sales
    FROM customers c
    JOIN orders o
        ON c.customer_id = o.customer_id
    GROUP BY
        c.customer_id,
        c.customer_name
)

SELECT
    customer_name,
    total_sales,
    RANK() OVER(ORDER BY total_sales DESC) AS sales_rank
FROM customer_sales
ORDER BY sales_rank;
```


Output:

Data Output Messages Notifications			
Showing rows: 1 to 793			
	customer_name character varying (150)	total_sales numeric	sales_rank bigint
1	Sean Miller	25043.050	1
2	Tamara Chand	19052.218	2
3	Raymond Buch	15117.339	3
4	Tom Ashbrook	14595.620	4
5	Adrian Barton	14473.571	5
6	Ken Lonsdale	14175.229	6
7	Sanjit Chand	14142.334	7
8	Hunter Lopez	12873.298	8
9	Sanjit Engle	12209.438	9
10	Christopher Conant	12129.072	10
11	Todd Sumrall	11891.751	11
12	Greg Tran	11820.120	12
13	Becky Martin	11789.630	13
14	Seth Vernon	11470.950	14
15	Caroline Jumper	11164.974	15
Total rows: 793		Query complete 00:00:00.578	

Insight: Combining joins, CTEs, and window functions provides a comprehensive customer performance report. The query identifies customer rankings while displaying their overall contribution to sales.

--Mini Project: Customer Sales Insights

--1. Who are the top 5 customers?

WITH customer_sales AS

```
(
    SELECT
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_name
)
```

```
SELECT *
FROM customer_sales
ORDER BY total_sales DESC
```

```
LIMIT 5;
```

Output:

	customer_name character varying (150) 	total_sales numeric 
1	Sean Miller	25043.050
2	Tamara Chand	19052.218
3	Raymond Buch	15117.339
4	Tom Ashbrook	14595.620
5	Adrian Barton	14473.571

Insight: The top five customers generate the highest revenue for the business.

Maintaining strong relationships with these customers can significantly impact future sales.

--2. Who are the bottom 5 customers?

```
WITH customer_sales AS
```

```
(
    SELECT
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_name
)
```

```
SELECT *
FROM customer_sales
ORDER BY total_sales ASC
LIMIT 5;
```

Output:

	customer_name character varying (150) 🔒	total_sales numeric 🔒
1	Thais Sissman	4.833
2	Lela Donovan	5.304
3	Carl Jackson	16.52
4	Mitch Gastineau	16.739
5	Roy Skaria	22.328

Insight: The bottom five customers contribute the least revenue. These customers may require targeted engagement strategies to improve purchasing activity.

```
--3. Which customers made only one order?
SELECT
    customer_id,
    customer_name,
    COUNT(DISTINCT order_id) AS orders_count
FROM superstore
GROUP BY customer_id, customer_name
HAVING COUNT(DISTINCT order_id) = 1;
```

Output:

	customer_id character varying (20) 🔒	customer_name character varying (150) 🔒	orders_count bigint 🔒
1	AO-10810	Anthony O'Donnell	1
2	AR-10570	Anemone Ratner	1
3	CJ-11875	Carl Jackson	1
4	JC-15385	Jenna Caffey	1
5	JR-15700	Jocasta Rupert	1
6	LD-16855	Lela Donovan	1
7	MG-18205	Mitch Gastineau	1
8	PH-18790	Patricia Hirasaki	1
9	RE-19405	Ricardo Emerson	1
10	RM-19750	Roland Murray	1
11	SM-20905	Susan MacKendrick	1
12	TC-21145	Theresa Coyne	1

Insight: These customers interacted with the business only once. Understanding why they did not return may help improve customer retention and repeat purchases.

--4. Which customers have above-average sales?

WITH customer_sales AS

```
(
    SELECT
        customer_name,
        SUM(sales) AS total_sales
    FROM superstore
    GROUP BY customer_name
)
```

```
SELECT *
FROM customer_sales
WHERE total_sales >
(
    SELECT AVG(total_sales)
    FROM customer_sales
);
```

Output:

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 294

	customer_name character varying (150) 🔒	total_sales numeric 🔒
1	Peter McVee	4115.662
2	Sandra Glassco	3406.581
3	Julie Kriz	2932.484
4	Alan Hwang	4805.344
5	Cindy Stewart	5690.055
6	Paul Prost	7252.608
7	John Stevenson	3868.020
8	Tamara Manning	2955.226
9	Dorothy Badders	3908.8028
10	Dionis Lloyd	3539.322
11	Jonathan Doherty	7610.864
12	Raymond Buch	15117.339
13	Resi Pölking	5153.084
14	Kristen Hastings	5990.799

Total rows: 294

Query complete 00:00:00.409

Insight: Customers exceeding the average sales threshold represent a high-performing customer segment. They offer opportunities for upselling, cross-selling, and loyalty programs.

--5. What is the highest order value per customer?

```
SELECT
    customer_id,
    customer_name,
    MAX(sales) AS highest_order_value
FROM superstore
GROUP BY customer_id, customer_name
ORDER BY highest_order_value DESC;
```

Insight: The highest order value indicates the maximum spending capacity of each customer. This information can help identify premium customers and purchasing trends.

Output :

Data Output				Messages	Notifications
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	customer_id character varying (20) 🔒	customer_name character varying (150) 🔒	highest_order_value numeric 🔒		
1	SM-20320	Sean Miller	22638.48		
2	TC-20980	Tamara Chand	17499.95		
3	RB-19360	Raymond Buch	13999.96		
4	TA-21385	Tom Ashbrook	11199.968		
5	HL-15040	Hunter Lopez	10499.97		
6	AB-10105	Adrian Barton	9892.74		
7	SC-20095	Sanjit Chand	9449.95		
8	BS-11365	Bill Shonely	9099.93		
9	SE-20110	Sanjit Engle	8749.95		
10	CC-12370	Christopher Conant	8399.976		
Total rows: 793		Query complete 00:00:00.307			